DATA ANALYTICS - PHASE 1

PULSEIQ

Phase 1 Review & Report

PREPARED BY

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Data Analytics Track

PHASE 1 PROJECT SUBMISSION · 2026

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Track: Data Analytics

Phase: Phase 1 — After Day 5
GitHub Repository: https://github.com/Aiswarya3006/PulseIQ
Git Tag: phase-1

1. Project Overview

PulseIQ is a data analytics project developed to process raw business data, validate data quality, create analytical metrics, and present useful insights through a dashboard. During Phase 1, I worked on ETL processing, warehouse validation, data-quality checks, metric views, and dashboard development.

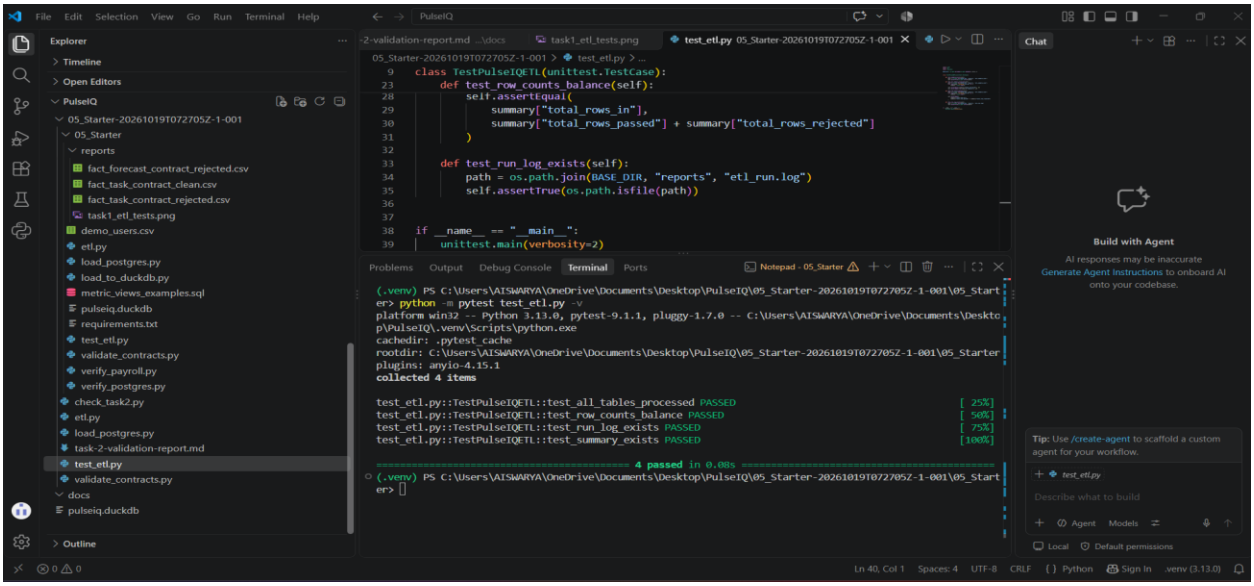
2. Work Completed in Each Task

Task 1 — ETL Pipeline and Automated Testing

I worked on processing raw data and preparing it for analytics. I ran automated tests to verify table processing, row-count balance, ETL run-log creation, and summary-file creation.

Result: All 4 automated tests passed.

Evidence: reports/task1_etl_tests.png



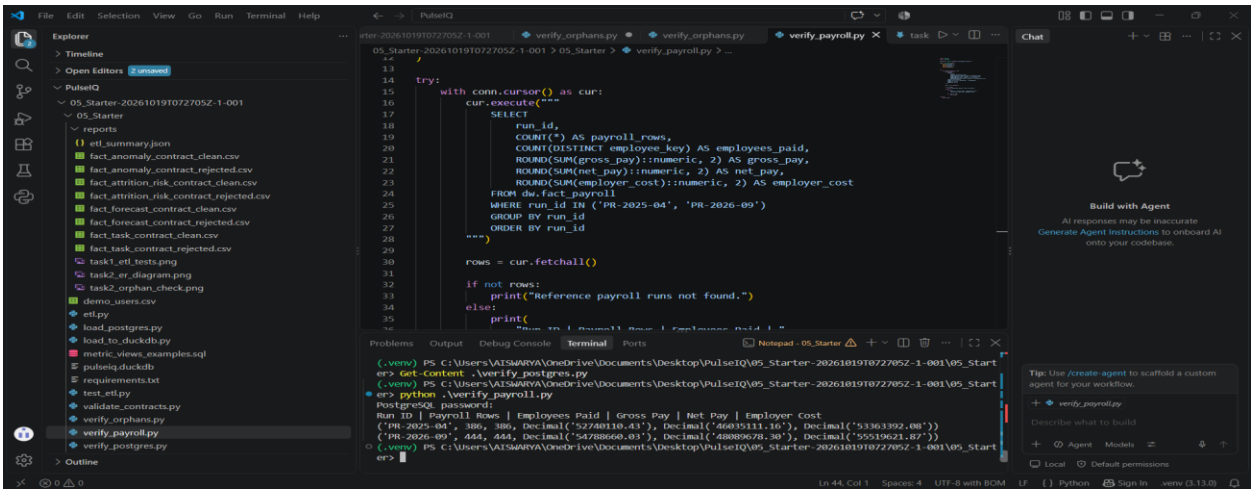
Task 2 — Data Warehouse and Validation

I verified the data warehouse structure and checked relationships between tables. I reviewed the ER diagram, checked for orphan records, and examined payroll reconciliation results.

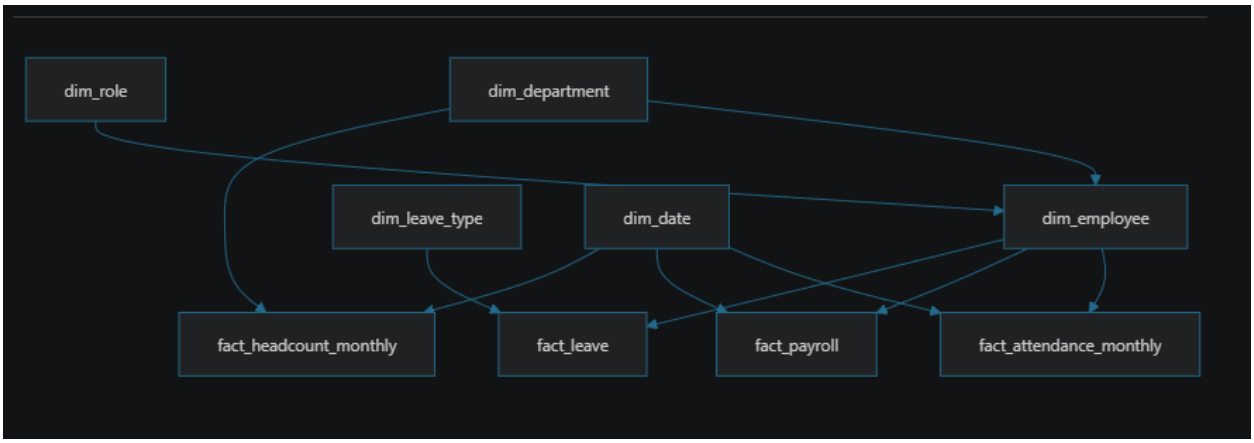
Result: The PostgreSQL warehouse contained 29 tables. The payroll verification reported 7,338 rows and 18 distinct payroll runs. The orphan check reported zero employee-key or department orphans in the checked PostgreSQL payroll data.

Evidence:

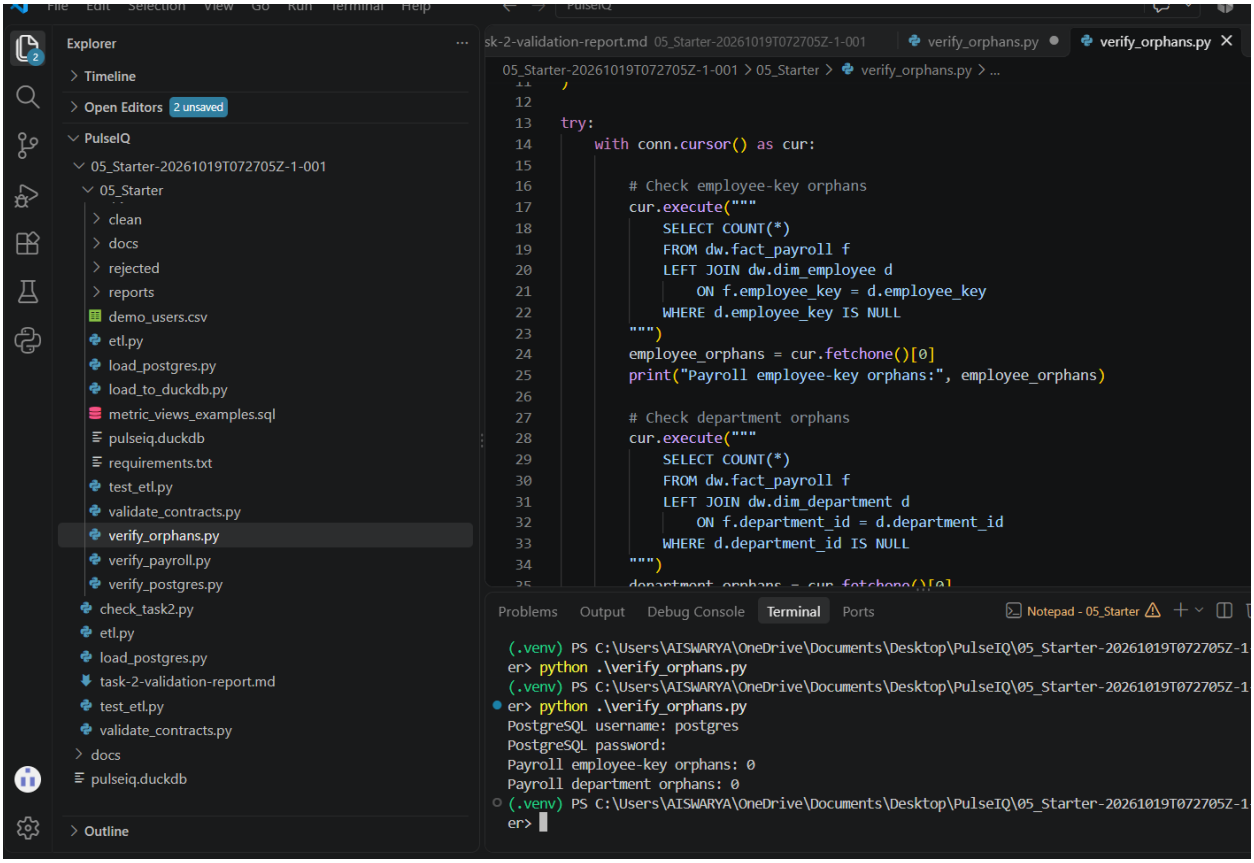
reports/task2_er_diagram.png:



reports/task2_orphan_check.png:



reports/task2_payroll_reconciliation.png:

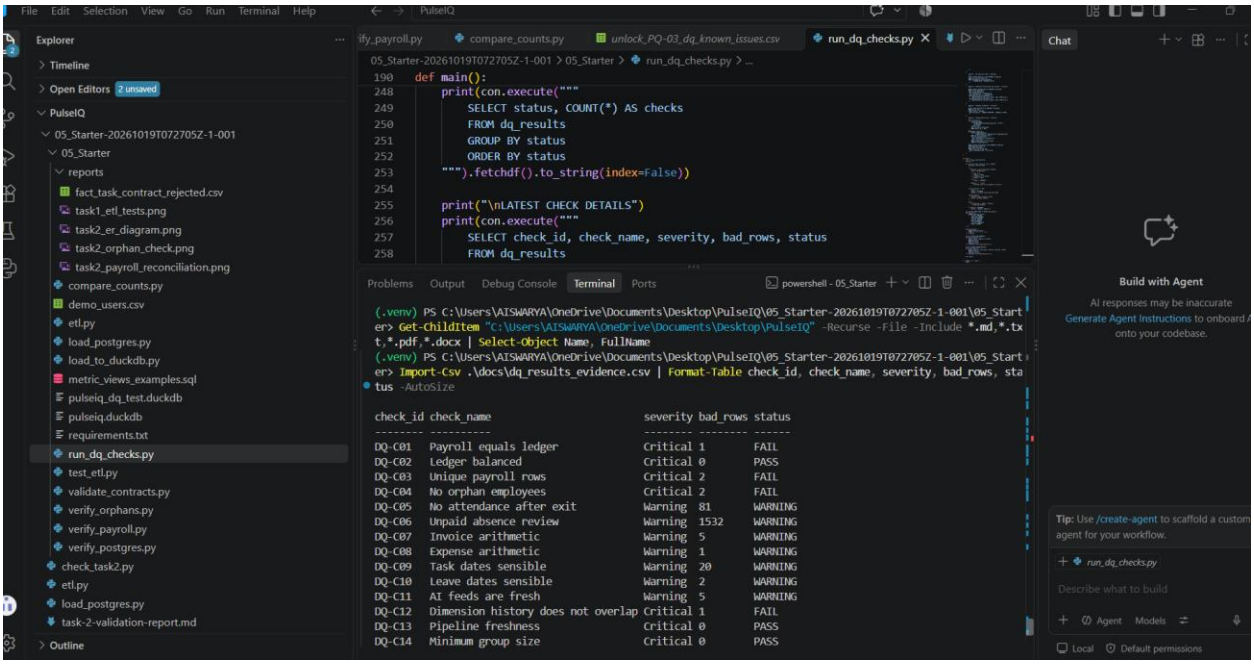


Task 3 — Data Quality Checks

I executed 14 data-quality checks to identify issues such as duplicate payroll records, missing employee references, payroll/ledger mismatches, invalid task and leave dates, invoice and expense issues, and overlapping employee-history periods.

Result: 3 checks passed, 4 checks reported failures, and 7 checks reported warnings. These results were recorded for review. Warnings require business-rule interpretation before deciding whether they represent actual defects.

Evidence: reports/Task3_Screenshot1_DQ_Results.png, docs/dq_missed_issues.md

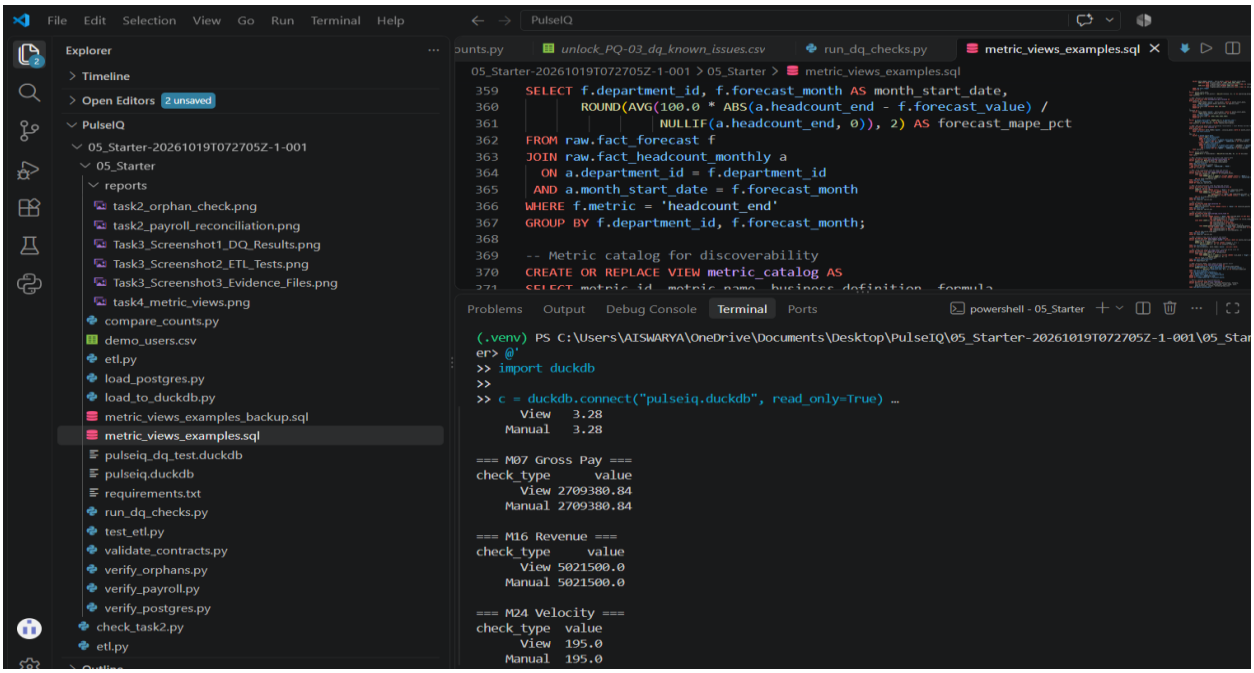


Task 4 — Metric Layer

I reviewed the metric views and metric dictionary used to define and calculate business measures. I also performed hand checks and captured evidence of the metric layer.

Result: Metric views m01 to m30 were reported present. The metric catalog contained 28 rows. The forecast MAPE view returned no rows because the forecast period did not overlap the available actual headcount period.

Evidence: reports/task4_metric_views.png, reports/task4_hand_checks.png, docs/metric-dictionary.md

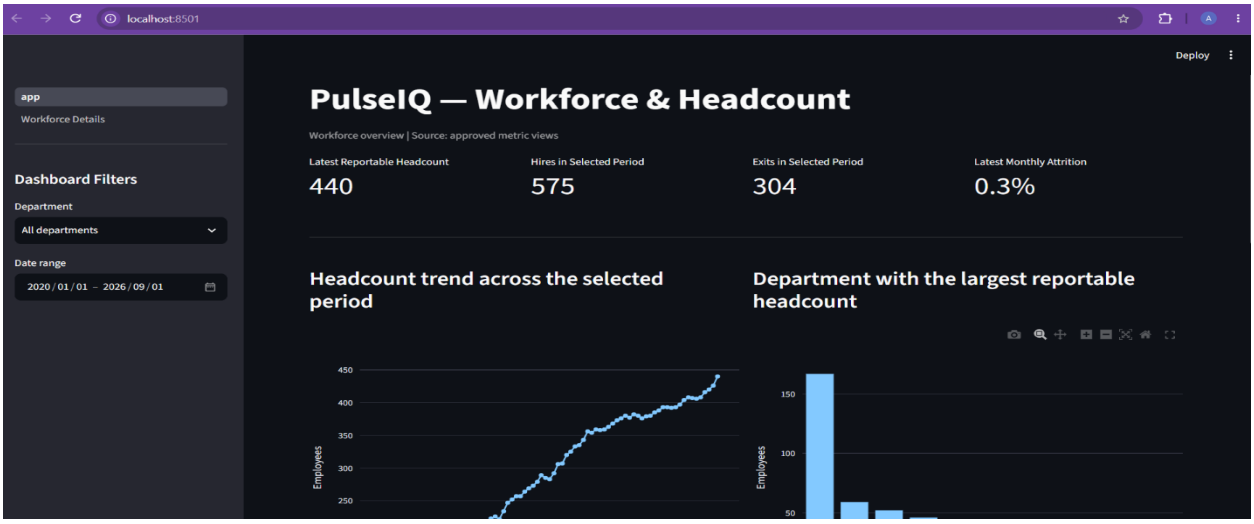


Task 5 — Dashboard Development

I worked on the PulseIQ dashboard to present business metrics and trends in a visual format. I saved a dashboard screenshot and committed the evidence to the GitHub repository.

Result: Dashboard evidence is available in the repository.

Evidence: reports/task5_dashboard.png



3. Overall Validation Results

- Automated ETL tests: 4 passed.
- Contract validation: 29,775 rows checked with zero validation errors; extra columns were flagged for review.
- Data-quality checks: 14 checks executed — 3 PASS, 4 FAIL, 7 WARNING.
- Metric views: m01 to m30 were reported present.
- Dashboard screenshot and supporting evidence were committed to GitHub.

These results represent the checks executed during Phase 1. Data-quality failures and warnings are documented rather than presented as successful checks.

4. What I Learned

I learned how to process and validate raw data using an ETL pipeline. I gained experience checking warehouse relationships, identifying data-quality issues, and reviewing metric calculations. I also learned how analytical metrics can be presented through a dashboard and why business context is important when interpreting data-quality warnings.

5. Problems Encountered and Solutions

During validation, some data-quality checks reported failures and warnings. I recorded the findings so they could be reviewed instead of changing the source data without justification. I also found that the forecast MAPE view had no overlapping actual periods, which prevented the metric from being calculated. Finally, I updated the phase-1 Git tag to point to the latest dashboard-evidence commit and pushed the tag to GitHub.

6. Plan for the Next Phase

In the next phase, I plan to review data-quality findings against business rules, validate forecast accuracy when suitable actual data becomes available, and practice creating a new metric from definition through dashboard presentation and manual verification within 45 minutes.

GitHub Repository

https://github.com/Aiswarya3006/PulseIQ